Nandhith Karthikeyan

B. Tech Robotics and AI, Amrita Vishwa Vidyapeetham Bengaluru Campus
nandhith.k@gmail.com | +91 74834 62090 | github.com/NK2511

Profile

Robotics and AI undergraduate with strong foundations in Reinforcement Learning and Mechanical Design. Demonstrated research capability through three accepted publications and hands-on experience developing humanoid locomotion at Xterra Robotics, IIT Kanpur. Passionate about bridging the gap between simulation and real-world hardware.

Education

B. Tech in Robotics and Artificial Intelligence| Amrita Vishwa Vidyapeetham, Bengaluru, **CGPA: 9.16** *[2023-Present]*
Leadership & Roles: Class Representative, Robocon Mechanical Design Lead, ACROM Club Executive

Skills

3D Modelling and Simulation: Fusion 360, RecurDyn, Mujoco, PyBullet, Stable Baselines3, OpenCV, mink (IK solver), MJ_Warp
Electrical: Microcontrollers (Arduino, ESP32), Raspberry Pi
Programming Languages: Python, C++, C#, MATLAB, ROS
Soft Skills: Team Management, Leadership Skills, Communication

Internships

IIT Kanpur & Xterra Robotics *[12/2025 – Present]*

- Authored the URDF model (12-DoF bipedal lower body) for the team's DRDO-funded Angad humanoid platform, defining inertial and collision properties for simulation stability
- Built a physics-based gait scaling algorithm minimizing Cost of Transport across joint work, motor heating, and impact losses
- Extended the RL training pipeline (Asimov, built on rsl_rl/MuJoCo/mjwarp) to imitate CoT-optimal gaits, achieving stable velocity tracking up to 2.5 m/s

Robocon Design Team Lead *[10/2024-06/2026]*

- Designed, modeled, and manufactured robust mechanical systems for autonomous and manual robots
- Led a multidisciplinary team of students, successfully qualifying for the Semifinals of Robocon 2025 and 2026.

Projects

Reinforcement Learning for 8 Degree of Freedom Quadruped gait *[07/2025 - 11/2025]*
Developed a complete URDF model of a quadruped and trained locomotion policies using Soft Actor-Critic in PyBullet, followed by deployment on hardware

Drone-Car Transformer *[08/2025 -11/2025]*
Ideated, designed and fabricated and innovative drone capable of seamlessly switching between aerial and ground-vehicle mode.

Turmeric Classification *[06/2025 - Present]*
Designed the end-to-end development of an automated turmeric classification system, gathering on-site data to train a robust deep learning model. Currently supplying real-time visual coordinate data to direct a delta robot for automated physical sorting.

Publications

An Inverted Compass Model for Passive Dynamic Brachiation *(Accepted in ISRM)*
Demonstrated existence of stable limit cycles for Passive Brachiation along a Slope

Five-bar Mechanism as Chess Playing Robot using Vision and AI *(Accepted in INaCOMM)*
Presented a low-cost, robotic chess-playing system that provides physical playing experience

Hybrid Method of PSO and L-BFGS-B for Path Planning for n-Serial Revolute Manipulator *(Accepted in IEEE)*
Proposed hybrid PSO-LBFGS method for path planning in redundant serial manipulators

Extra-Curricular Interests

12 years of experience in Carnatic Violin